Data Visualization

176 terms · name, then the phrase to paste · Workbook Library

Compare & change

Column chart a column chart with one bar per period, starting at zero, bars a little wider than the gaps between them, light horizontal gridlines and no border around the plot

Sorted bar chart a horizontal bar chart sorted from largest to smallest, category labels left of the bars, the value printed at the end of each bar instead of an x-axis

Grouped bar chart a grouped bar chart with the series side by side inside each category, at most four series, a consistent color per series and a legend placed above the plot

Stacked bar chart a stacked bar chart where the segments sum to each bar's total, the most important series at the bottom on the baseline, no more than four segments, and the total labeled above each bar

Lollipop chart a horizontal lollipop chart sorted by value: a thin stem from the zero baseline to a solid dot, labels on the left, values next to each dot

Dot plot a Cleveland dot plot: one row per category, a dot for each series on a shared horizontal axis, faint dotted guide lines across each row, rows sorted by the latest value

Dumbbell chart a dumbbell chart: for each category, two colored dots (before and after) joined by a thick gray bar, rows sorted by the size of the gap, and a two-item legend above

Slope chart a slope chart comparing two years: one line per category from the left axis to the right axis, labels with values at both ends, the steepest riser in color and the rest in gray

Bullet chart bullet charts, one per KPI: a dark bar for the actual value, a short vertical line for the target, and three shades of one hue behind it for poor, satisfactory and good ranges

Radar chart a radar chart with six axes scored 0 to 1, two series drawn as translucent filled shapes with solid outlines, light concentric gridlines and axis labels outside the rim

Pictogram chart a pictogram chart where each icon stands for a fixed count, icons in tidy rows of the same size, partial icons cut off for remainders, and a key that says what one icon equals

Line chart a line chart with time on the x-axis, a 2px line per series, no markers on every point, light horizontal gridlines, and each line labeled at its right end instead of a legend

Area chart an area chart of one series with a solid line on top of a soft translucent fill down to a zero baseline, light gridlines and the latest value labeled at the end

Stacked area chart a stacked area chart with the layers summing to the total, the most stable series at the bottom, thin white seams between layers, and labels placed inside the widest part of each layer

Streamgraph a streamgraph: smooth stacked layers centered around a flowing midline, no y-axis, each layer in its own color with its label placed on the layer at its widest point

Step chart a step chart where the line stays flat until each change and then jumps vertically, with each step labeled with its new value and date

Sparklines sparklines in a table: one small line per row with no axes or gridlines, the last value marked with a dot and printed to the right, and the high point marked in gray

Candlestick chart a candlestick chart: a thin wick from high to low and a body from open to close for each day, up days in green and down days in red, price axis on the right

Horizon chart horizon charts, one thin row per series: values above the baseline in layered blues and below in layered reds, darker shades for higher bands, all rows sharing one time axis

Bump chart a bump chart where y is rank with first place at the top, thick colored lines with a dot at each period, and names labeled at both ends

Calendar heatmap a calendar heatmap: one square per day in columns by week and rows by weekday, five shades of one hue from light (none) to dark (most), month labels on top and a small legend

Pie chart a pie chart with at most five slices, starting at 12 o'clock and sorted clockwise from largest, labels with percentages placed next to each slice instead of a legend

Donut chart a donut chart with a thick ring, at most four segments, the total or key percentage in large type in the center hole, and short labels outside the ring

100% stacked bar a 100% stacked bar chart where each bar sums to 100%, segments in a consistent order with the most important one on the baseline, percentages labeled inside large segments

Treemap a treemap with rectangles sized by value and grouped by parent category, each group in its own hue with children in the same hue, labels inside every rectangle large enough to hold one

Sunburst chart a sunburst chart: the inner ring for top-level categories, the outer ring for their children in a lighter tint of the parent's color, and a total in the center

Icicle chart an icicle chart: the whole on the top row, each level below split proportionally by its children, rectangles colored by top-level parent and labeled where they fit

Waffle chart a waffle chart of 100 small squares in a 10 by 10 grid, each square one percent, filled by category from the bottom left, with a legend that lists each share

Marimekko chart a Marimekko chart where column width is each market's share of the total and the stacked segments are each company's share within that market, all tiles labeled with percentages

Parliament chart a parliament chart: one dot per seat arranged in concentric half-circle rows, grouped by party from left to right, with the seat total in the middle

Spread, links & maps

Histogram a histogram with equal-width bins and touching bars, a count axis starting at zero, and the bin edges labeled on the x-axis; mention the bin width in the subtitle

Box plot box plots, one per group on a shared axis: a box for the middle 50%, a bold median line, whiskers to 1.5 times the IQR, and outliers as small open circles

Violin plot violin plots, one per group: a mirrored smoothed density shape filled with the group color, a short bold line at the median, and all violins on one shared value axis

Strip plot a strip plot with every observation as a small semi-transparent dot, jittered horizontally within each group, and a short line marking each group's median

Beeswarm plot a beeswarm plot: one dot per item along a single axis, dots packed without overlap so the swarm's silhouette shows the distribution, a few notable dots labeled

Ridgeline plot a ridgeline plot: one smoothed density curve per group stacked vertically with slight overlap, filled with a sequential palette from light to dark, labels on the left

Density curve overlaid density curves, one per group, each a solid line over a translucent fill of the same color, labeled at its peak instead of with a legend

Cumulative distribution an empirical cumulative distribution plot: a step line per group from 0% to 100%, a percentage y-axis, and dashed guides marking the median and 90th percentile

Scatter plot a scatter plot with one semi-transparent dot per item, both axes labeled with units, a thin fitted trend line, and the three most extreme outliers labeled

Bubble chart a bubble chart: x and y for two measures, bubble area proportional to a third value, color by group, white outlines so overlapping bubbles stay readable, and a size key

Connected scatter plot a connected scatter plot: x and y for two measures, points joined in time order by a smooth line, the first and last points and key years labeled

Heatmap a heatmap with rows and columns as two categories, each cell filled from a single-hue sequential scale, thin gaps between cells, and a color legend with labeled stops

Hexbin plot a hexbin plot: points binned into small hexagons colored by count on a sequential scale, empty hexagons left blank, and a legend for the counts

Parallel coordinates a parallel coordinates plot with one vertical axis per attribute, one thin translucent line per item colored by group, and one selected item drawn thicker

Scatterplot matrix a scatterplot matrix: a small scatter plot for every pair of variables arranged in a grid, variable names on the diagonal, shared axes along each row and column

Quadrant chart a quadrant chart: a scatter plot divided by dashed lines at the midpoint of each axis, each quadrant labeled in its corner with a name, and the items labeled next to their dots

Sankey diagram a Sankey diagram with stages left to right, flow bands sized by volume and colored by their source, narrow vertical node bars, and labels with totals next to each node

Chord diagram a chord diagram: groups as colored arcs around a circle, translucent ribbons between them sized by flow and colored by the larger source, group names outside the circle

Network graph a force-directed network graph: nodes as circles sized by number of connections and colored by community, thin gray links, and only the biggest hubs labeled

Arc diagram an arc diagram: nodes in a row along a horizontal baseline, links drawn as semicircular arcs above, arc thickness by link weight, and one selected node's arcs highlighted

Tree diagram a tree diagram from a root at the top to leaves at the bottom, right-angled elbow connectors, nodes colored by level, and every node labeled

Gantt chart a Gantt chart: one row per task with a rounded bar from start to end date on a weekly axis, bars colored by phase, and a vertical red line marking today

Timeline a horizontal timeline with a single axis, a dot for each event, labels alternating above and below the line with thin leader lines, and year ticks along the axis

Funnel chart a funnel chart: one centered bar per stage sized by count, stage names and counts on the left, and the conversion rate from the previous stage on the right

Choropleth map a choropleth map with regions filled from a five-step sequential scale of one hue, values normalized per capita, thin white borders, and a legend with labeled class breaks

Proportional symbol map a proportional symbol map: translucent circles with area proportional to the count, placed on each city, biggest drawn first so small ones stay on top, and a size legend

Dot density map a dot density map where each tiny dot equals a fixed number of people, dots colored by group, a pale base map with no borders, and a key stating what one dot equals

Tile grid map a tile grid map with one equal-size square per region arranged to approximate geography, filled on a sequential scale, each tile labeled with its short code

Cartogram a Dorling cartogram: each region as a circle sized by population and nudged so circles do not overlap, keeping rough geographic positions, colored by category

Flow map a flow map: curved arcs from one hub to each destination, arc thickness proportional to volume, a muted base map, and the hub marked with a dark dot

Spike map a spike map: thin translucent triangles rising from each location with height proportional to the value, over a pale base map

Encodings & scales

Position on a common scale encode the key measure as position along a shared axis, the most accurate channel, rather than as color or size

Length show amounts as bar length from a shared zero baseline so lengths compare directly

Angle and arc use angle (pie or donut slices) only for two or three shares with large differences; use bar length when values are close

Area encode magnitude as circle area (radius proportional to the square root of the value) and include a size legend with three reference circles

Color lightness encode the value as lightness of a single hue, light for low and dark for high, with a legend showing the steps

Color hue use hue only for categories, at most six distinct hues, with every other category in neutral gray

Shape use marker shape (circle, triangle, square) as a redundant second channel for groups so the chart still works in grayscale

Line style draw measured data as solid lines and forecasts, targets or estimates as dashed lines in the same color

Transparency draw points at about 20% opacity so overlapping points build up into darker areas where data is dense

Zero baseline start the value axis at zero for bars and areas, and draw the zero line slightly darker than the other gridlines

Log scale use a logarithmic y-axis with ticks at 1, 10, 100, 1,000 and 10,000 and the words "log scale" in the axis title, so constant growth rates appear as straight lines

Indexed to 100 index every series to 100 at the start date so lines begin at the same point, with a reference line at 100 and the y-axis labeled "index, Jan = 100"

Change from a baseline plot year-over-year percent change as bars above and below a zero line, positive in blue and negative in red, with the zero line drawn in dark ink

Per capita normalize counts by population and show rates per 100,000 people, stating the denominator in the subtitle

Shared axis across panels give every small-multiple panel the same y-axis range so heights compare across panels; only free the axes if each panel's shape matters more than its size

Dual-axis chart Avoid a dual-axis chart with two unrelated y-scales; instead use two aligned panels sharing one x-axis, or index both series to a common start

Truncated bar axis Avoid a bar chart whose axis starts above zero; start bars at zero, or use dots on a zoomed axis if small differences matter

Inverted or unlabeled axis Avoid an inverted y-axis where higher values sit lower; keep larger values higher and label every axis

Moving average plot a 7-day moving average as a bold line over the raw daily values drawn as faint thin bars or a light line, and say "7-day average" in the label

Running total show a cumulative year-to-date line against a straight dashed line toward the annual target, labeling the gap at the latest point

Median versus mean summarize with the median rather than the mean because the data is skewed, mark both on the histogram, and state which one the headline uses

Smoothed trend line add a smoothed LOESS trend curve through the scatter in dark ink, keep the points visible behind it, and avoid extending the curve beyond the data

Forecast with uncertainty band show the forecast as a dashed line continuing the actuals, inside a shaded band for the 80% interval that widens over time, and label the band

Error bars draw each estimate as a dot with a horizontal 95% confidence interval line, sorted by estimate, and note the interval in the subtitle

Top N plus "Other" show the top five categories sorted by value and group the remainder into a single gray "Other" bar placed last, regardless of its size

This year against last year overlay each year as a separate line on a shared January to December axis, previous years in thin gray, the current year in bold color and labeled

Color for data

Categorical palette a categorical palette of up to six distinct hues with similar lightness, the same color for each group on every chart, and gray for anything else

Sequential palette a single-hue sequential palette from very light for low values to dark for high values, in five to seven steps

Diverging palette a diverging palette from red through a pale neutral to blue, centered exactly on the meaningful midpoint, with equal steps on both sides

Gray plus one accent draw every series in light gray except the one the story is about, which gets a single strong accent color and a direct label

Stepped classes bin values into five classes with round-number breaks and a stepped legend, rather than a continuous gradient

Bivariate color a bivariate choropleth with a 3 by 3 blended palette and a square legend that labels both axes with arrows for "higher"

Meaningful colors use conventional, meaningful colors (blue for gains and red for losses, party colors for parties) and repeat the meaning in labels or icons

Cyclic palette a cyclic color palette for the hour of day so 11pm and midnight are similar colors, shown with a circular legend

Viridis the viridis colormap (dark purple to yellow), perceptually uniform and color-blind safe

Magma and inferno the magma colormap from near-black through magenta and orange to pale yellow

Cividis the cividis colormap (navy to yellow), designed to look the same with or without color vision deficiency

ColorBrewer Blues the ColorBrewer "Blues" sequential palette in five to seven classes

ColorBrewer RdBu the ColorBrewer "RdBu" diverging palette with white at the midpoint

Okabe-Ito the Okabe-Ito color-blind-safe categorical palette (orange, sky blue, bluish green, yellow, blue, vermillion, reddish purple)

Tableau 10 the Tableau 10 categorical palette: muted, evenly weighted hues

Grayscale with one spot color only black, three grays and one accent color, so the chart still works when printed in black and white

Dark-mode palette a dark-mode version of the palette: the same hues lifted in lightness and slightly desaturated, on a near-black background with very faint gridlines

Rainbow scale Avoid a rainbow or jet colormap for ordered values; use a perceptually uniform palette like viridis instead

Red against green Avoid red versus green as the only cue; use a blue-orange pairing, or add icons, labels or patterns

Too many colors Avoid more than six category colors; highlight the few that matter and put the rest in gray

Dashboards & interaction

KPI tiles a row of four KPI tiles across the top, each with a short label, the current value in large type, the change versus last period in green or red with an arrow, and a sparkline

Inverted pyramid lay the dashboard out as an inverted pyramid: KPI tiles on top, two or three trend and breakdown charts in the middle, and a detail table at the bottom

Small multiples small multiples: one small chart per group in a grid, identical axes and scales, the group name as each panel's title, and the other groups drawn faintly in gray behind each panel for context

One hero chart one large hero chart taking about two-thirds of the layout, with three small supporting tiles stacked beside it

Filter bar a filter bar across the top with a date range, region dropdown and segment toggle, applying to every chart, with active filters shown as removable chips

Table with in-cell bars a table with right-aligned tabular numbers, a small horizontal bar inside each value cell scaled to the column maximum, and one column shaded as a heatmap

Status board a status board listing each item with a pill reading "At risk", "Watch" or "On track" in red, amber or green, sorted worst first, with a one-line reason

Annotated report page a one-page static report: headline finding at the top, one fully annotated chart, a short paragraph of explanation, then two supporting charts, with all values labeled because there is no hover

Tooltip tooltips on hover and tap showing the item name in bold and its values with units, positioned next to the point and never covering it

Hover to highlight on hover, highlight the series under the cursor in its color with a label, and fade all other series to light gray

Crosshair readout a vertical crosshair that follows the cursor across the time axis, showing the date and each series' value at that point in a compact readout, synced across all charts

Linked brushing linked brushing: dragging a rectangle in the scatter plot selects those items, which stay colored in every other chart while unselected items fade to gray

Drill-down clicking a bar drills into its breakdown with an animated transition, and a breadcrumb above the chart ("All regions › Europe › France") leads back up

Overview plus detail a large detail chart above a small overview strip of the full time range, with a draggable, resizable window on the strip that sets the detail view's range

Interactive legend make the legend interactive: clicking an item toggles that series, double-clicking isolates it, and hidden items stay in the legend in gray with a strike-through

Animated transitions animate changes: when filters or sorting change, bars and points move smoothly to their new positions over about 400ms so each item can be followed

Scrollytelling scrollytelling: a sticky chart beside or behind scrolling text cards, each card triggering one change in the chart (highlight, zoom, add a series) with a smooth transition

Animated time slider a time slider with a play button that animates the chart through each year, the current year shown large and faint behind the data

Annotation & story

Headline title a headline title that states the main finding as a short sentence, with a smaller subtitle naming the measure, units and time period

Direct labels label each series directly at its end in the series color instead of using a legend

Callout note add one or two short callout notes in small text with thin leader lines pointing to the exact data points they explain

Reference line add a labeled dashed reference line for the target value across the plot, and color bars that miss the target differently

Shaded period shade the relevant period with a light gray vertical band behind the data, labeled at the top

Event markers mark key events with thin vertical lines and short labels at the top of the chart

Units on the top tick put the units on the top y-axis label (for example "60k cups") instead of a rotated axis title

Source and notes line a small footer under the chart with "Source:" and the data source, a short note on any adjustments, and the date of the data

Plain-language uncertainty state uncertainty in plain words next to the key number (for example "likely between 41% and 47%") in addition to any band or error bar

Martini glass structure it as a martini glass: three or four guided steps that make the main point in order, then an open, explorable chart with filters at the end

Build it up step by step reveal the chart in three steps: first the axes and one series with the setup sentence, then the comparison series, then the highlight and annotation that delivers the finding

Before and after show before and after side by side with identical scales and layout, labeled "Before" and "After", with the key difference annotated

Zoom from context to detail show the full context chart first, then a zoomed inset of the important range, joined to it by guide lines so the reader sees where the detail comes from

One big number lead with one big number in very large type, a one-line explanation beneath it, and a small supporting chart for context

Make the number concrete translate the key number into a familiar comparison (for example "enough to fill 40 Olympic pools") and show it as a simple row of icons

How to read this chart add a small "How to read this" key that explains one example mark with arrows and plain words

You are here let the reader enter their own value and highlight their position in the distribution with a "You" label, everyone else in gray

3D effects Avoid 3D effects, perspective or extrusion of any kind; keep every chart flat and 2D

Pie with many slices Avoid pie charts with more than five slices; use a sorted bar chart instead

Spaghetti chart Avoid line charts with more than five colored lines; highlight the important few and gray the rest, or split into small multiples

Chartjunk Avoid chartjunk: gradients, shadows, heavy borders, background images and dense gridlines; keep only the data, labels and faint gridlines

Rotated labels Avoid rotated or diagonal axis labels; switch to horizontal bars so labels read normally

Legend far from the data Avoid a separate legend far from the data; label series directly next to the lines or bars

Styles & landmarks

Tufte minimalism Tufte-style minimalism: maximize the data-ink ratio, no gridlines or chart border, range-frame axes that span only the data, small serif labels, thin dark lines and direct labels

Weekly news magazine news-magazine chart style: a small red rectangle above a bold sans-serif title and a short subtitle, white background, light horizontal gridlines only, y-axis labels on the right, flat dark-blue and light-blue lines, source line at bottom left

Financial paper financial-newspaper chart style: warm salmon-pink background, a serif headline, slim lines in deep blue, claret and teal, fine horizontal gridlines, labels in a clean sans-serif

Newspaper graphics desk newspaper graphics style: white background, data mostly in light gray with one series in a strong color, short annotations in small text placed directly on the chart, very light gridlines and no legend

Data-science blog data-blog chart style: light gray background, prominent white gridlines, thick bright lines in blue, red-orange and gold, a bold left-aligned title and subtitle, no axis spines

Bold black business bold dark editorial style: black background, very large bold white sans-serif headline, saturated electric blue, magenta and orange data, thin dark-gray gridlines

Research explorer research-explorer chart style: white background, title and explanatory subtitle at the top left, muted multi-color lines labeled at their ends, dotted light gridlines, and a source and license line at the bottom

Hand-drawn comic hand-drawn comic style: wobbly black ink lines, handwritten labels, no gridlines, a few arrows with casual annotations, on a white background

Swiss modernist Swiss modernist style: strict grid, a neo-grotesque sans-serif in two weights, black on white with one red accent, flush-left text, large numerals and no decoration

Terminal chart a terminal-style text chart: horizontal bars drawn with block characters (▇▇▇▇) in a monospaced font, values printed after each bar, green on a near-black background

Playfair's engraved charts in the style of an 18th-century engraved statistical plate: cream paper, fine sepia grid, two hand-drawn time series with the gap between them washed in pale yellow and pink, copperplate script labels

Nightingale's rose a Nightingale-style polar area diagram: twelve monthly wedges radiating from the center with area proportional to value, layered pale blue, red and black, on cream paper with fine serif labels

Minard's flow map a Minard-style flow map: a band whose width is the quantity, tan for the outward journey and black for the return, drawn over a sparse map with a linked temperature line underneath

Snow's cholera map a John Snow-style dot map: a fine black street plan with each case drawn as a small black tick stacked at its address, clustering around a marked source, on aged paper

Du Bois data portraits in the style of W.E.B. Du Bois's 1900 data portraits: flat gouache colors (crimson, gold, emerald green, black, tan) on buff paper, hand-lettered all-caps titles, bold simple forms with thin black outlines

Isotype Isotype style: rows of identical flat pictograms, each standing for a fixed quantity, in two or three solid colors (black, red, blue) on off-white, with plain sans-serif labels

Bertin's visual variables in the manner of Jacques Bertin: black ink on white, a reorderable matrix whose cells vary only in mark size and value, with a strict grid and small sans-serif labels

Gapminder bubbles a Gapminder-style bubble chart: income on a log x-axis, life expectancy on the y-axis, bubbles sized by population and colored by world region (cyan, pink, lime, yellow) with dark outlines, and the year as a huge pale numeral in the background

Pulsar plot a pulsar-style ridgeline: many thin white line traces stacked closely on a black background, each filled black so it hides the lines behind, with sharp peaks clustered in the center

Data humanism data humanism style: each record drawn as a small hand-made glyph whose petals, colors and marks encode its attributes, soft watercolor tones and fine ink lines on white, with an illustrated legend

Generative text drawings a generative drawing of a text: each sentence a straight segment whose length is its word count, turning right at every sentence break, colored by theme, on white with no axes

Delicate radial data art delicate data art: hair-thin lines and small circles in a radial arrangement, a soft palette of dusty pink, sage and ochre on off-white, tiny italic labels and generous empty space

Luminous radial luminous radial data art: arcs and circles arranged around a center, vivid magenta, orange and teal gradients with a soft glow on a deep navy background, minimal thin labels

Emotional arcs emotional arc visualization: each person drawn as an arc in bright orange that stops where their life ended, continuing as a faint gray ghost for the years they lost, on a dark charcoal background

Illustrated charts an illustrated chart that looks hand-painted: bars drawn as real objects with uneven watercolor fills, handwritten labels and values, warm muted colors on a cream background

Climate stripes climate-stripes style: one thin vertical stripe per year, colored on a red-blue diverging scale centered on the long-term average, with no axes, labels or gridlines

Generative data art generative data art: thousands of thin flowing strokes following a flow field driven by the data, luminous color gradient from deep blue through violet to warm orange, dark background, no axes

Sports analytics chart a sports-analytics shot chart: a faint half-court outline, hexagons sized by number of attempts and colored from cool blue (below average) to hot red (above average), with a compact legend